

Bud seed cues comparison model outlines

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1 Methods

1. We used different data and models for budburst and seed germination.
 - (a) We have complete data for 191 species for the budburst model, this data following the code used in MoralesCastilla *et al.* 2024 publication, including removal of outlier species and correcting *Acer pseudolatauns* for *Acer pseudoplatanus*.
 - (b) We have complete data for 329 angiosperm species and 107 gymnosperm species for the seed germination model.
2. We modeled budburst as a function of forcing, chilling, and photoperiod, accounting for phylogeny:

$$\hat{y}_i = \alpha_{sp[i]} + \beta_{force,sp[i]} \cdot forcing_i + \beta_{chill,sp[i]} \cdot chilling_i + \beta_{photo,sp[i]} \cdot photoperiod_i$$

where $sp[i]$ is the species of observation i , and species-level intercepts and slopes (α_{sp} , $\beta_{force,sp}$, $\beta_{chill,sp}$, $\beta_{photo,sp}$) are modeled according to species phylogenetic relationships.

3. We ran the model with forcing and chilling for seed germination considering phylogeny in the model
4. We modeled seed germination as a function of forcing and chilling, accounting for phylogeny, following the same structure as above (excluding the photoperiod term). The response variable was standardized to the $[0, 1]$: for germination measured as a percentage, we used the proportion directly; for germination recorded as a binary outcome, we coded observations as germinated (1) or not (0).
5. After fitting both models, we extracted the species-level parameter estimates and compared the chilling and forcing effects on budburst versus seed germination for the 38 species overlapping between the two datasets.